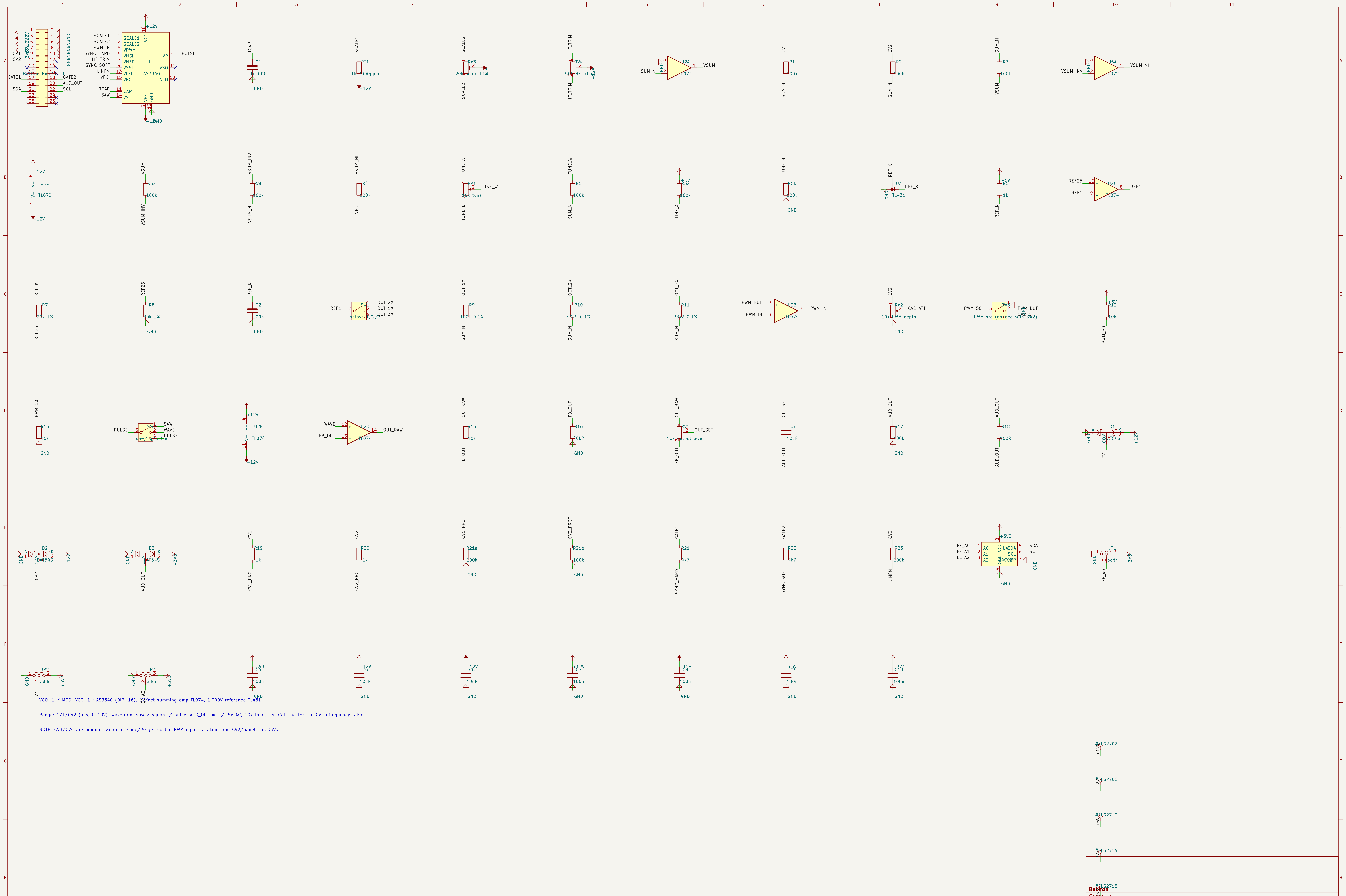
J1
Bukifon Bus 26 pin
CV1
CV2
GATE1
GATE2
AUD_OUT
SDA
SCL
+12V
-12V
+5V
+3V3
GND
U1
AS3340
SCALE1
SCALE2
PWM_IN
SYNC_HARD
HF_TRIM
SYNC_SOFT
LINFM
VFCI
TCAP
SAW
VPWM
VHSI
VHFT
VSSI
VLFI
CAP
VS
VCC
VEE
VP
VSO
VTO
PULSE
C1
1n C0G
RT1
1k 3300ppm
RV3
20k scale trim
RV4
50k HF trim
U2A
TL074
SUM_N
VSUM
R1
R2
R3
100k
U5A
TL072
VSUM_INV
VSUM_NI
U5C
R3a
R3b
R4
VFCI
RV1
10k tune
TUNE_A
TUNE_B
TUNE_W
R5
R5a
R5b
U3
TL431
REF_K
R6
1k
U2C
REF25
REF1
R7
20k 1%
R8
30k 1%
C2
100n
SW1
octave 1/2/3
OCT_1X
OCT_2X
OCT_3X
R9
100k 0.1%
R10
49k9 0.1%
R11
33k2 0.1%
U2B
PWM_BUF
RV2
10k PWM depth
CV2_ATT
SW3
PWM src (ganged with SW2)
PWM_50
R12
10k
R13
SW2
saw/sq/pulse
SAW
WAVE
U2E
U2D
FB_OUT
OUT_RAW
R15
R16
40k2
RV5
10k output level
OUT_SET
C3
10uF
R17
R18
100R
D1
D2
D3
BAT54S
A
K
COM
R19
R20
CV1_PROT
CV2_PROT
R21a
R21b
R21
R22
4k7
R23
LINFM
U4
24C02
A0
A1
A2
EE_A0
EE_A1
EE_A2
WP
JP1
JP2
JP3
addr
C4
C5
C6
C7
C8
C9
C10
#FLG2702
#FLG2706
#FLG2710
#FLG2714
#FLG2718

VCO-1 / MOD-VCO-1 : AS3340 (DIP-16), 1V/oct summing amp TL074, 1.000V reference TL431.

Range: CV1/CV2 (bus, 0..10V). Waveform: saw / square / pulse. AUD_OUT = +/-5V AC, 10k load, see Calc.md for the CV->frequency table.

NOTE: CV3/CV4 are module->core in spec/20 §7, so the PWM input is taken from CV2/panel, not CV3.

**Bukifon**
Sheet: /
File: MOD-VCO-1.kicad_sch

***Title: MOD-VCO-1***

Size: A2
Date:
**Rev: 1.0**
KiCad E.D.A. 10.0.4
Id: 1/1